Enhancing Accessibility and Reproducibility of Flow Cytometry Data Analysis with Panel-Agnostic Machine Learning-Based Automated Cross-Panel Sample-Level Classification for Acute Leukemia

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Introduction

- The lack of standardization in panel design and the complexity of manual analysis for flow cytometry (FC) data have hindered the speed and the spread of both implementation and the service capacity of flow cytometry being in the emerging countries.
- This study employs multiple machine learning (ML) approaches to develop an automated **panel-agnostic AML classification at both sample and cell levels** from several distinct flow cytometry laboratories.

Experiment A: Cross-Panel Sample Level Classification

Sample Level Dataset

A total of 220 newly-diagnostic FC records (110 AML and 110 non-neoplastic) were collected from five sites (three from Taiwan and two from US). Each laboratory applied different panels for AML and non-neopolastic diagnosis. 10 parameters were found commonly adapted in these flow panels.

Dataset	VGH	NTUH	RPCCC	UPMC	NTUCC	All Sites
Flow Panel	Euroflow AML/MDS	NTUH Diagnostic	ClearLLab 10C	UPMC Diagnostic	Euroflow AML/MDS	4 panels
Parameters (n)	18	19	31	30	11	10 in common*
Sample Number (AML, non-neoplastic)	17, 18	25, 22	27, 26	24, 25	17, 19	110, 110
Specimen Number (PB, BM)	0, 35	0, 47	5, 48	0, 49	0, 36	5, 215

* Common parameters: FSC-A, FSC-H, SSC-A, CD7, CD19, CD34, CD45, CD56, CD117, HLA-DR
Abbreviation: VGH: Taichung Veterans General Hospital, NTUH: National Taiwan University Hospital, RPCCC: Roswell Park Comprehensive Cancer Center, UPMC: University of Pittsburgh Medical Center, NTUCC: National Taiwan University Cancer Center

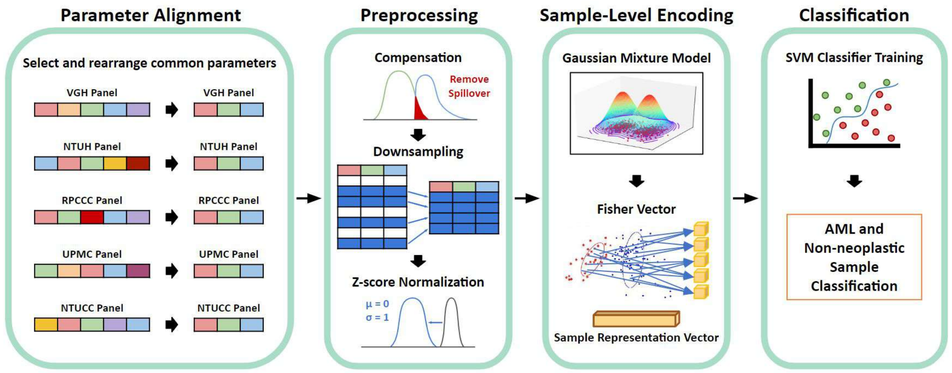
Sample Level Classification Results

The sample level classification models trained with 10 common parameters achieved comparable performance with an AUC of 99.32% compared to models trained with site-dependent parameters. (AUC ranging from 99.48% to 100%)

Dataset	VGH	NTUH	RPCCC	UPMC	NTUCC	All Sites
Parameters (n)	18	19	31	30	11	**10 in comon**
AUC	100.00%	100.00%	100.00%	99.48%	100.00%	**99.32%**
Accuracy	96.97%	100.00%	96.30%	97.92%	94.44%	**99.09%**
Sensitivity	93.33%	100.00%	92.59%	100.00%	88.89%	**99.10%**
Specificity	100.00%	100.00%	100.00%	95.83%	100.00%	**99.10%**

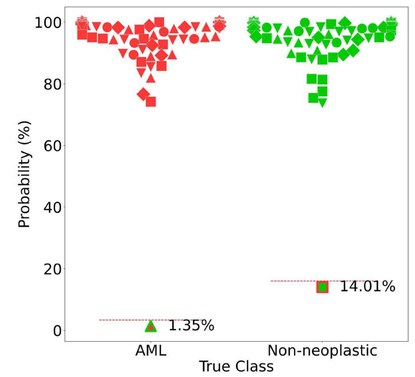
Abbreviation: AUC: Area Under the receiver operating characteristic (ROC) Curve

Sample Level Classification Framework

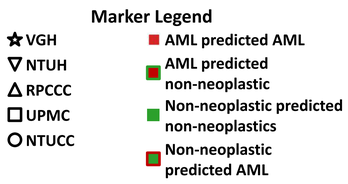


1. Select and rearrange common parameters across different panels
2. Pre-process data with compensation, downsampling (ratio=0.2), and sample-wise z-score normalization
3. Utilize Gaussian Mixture Model (GMM) and Fisher Vector to transform each sample's data into a 1D representation vector
4. Apply 3-fold cross validation to train and validate a Support Vecotr Macine (SVM) for binary classification

Ranking of Sample Prediction Probability of All Sites



This visualization tool ranks the sample prediction probabilities from small to large. The red line in each class represents the probability threshold that our method mis-classifies samples.



Experiment B: Cross-Panel Cell Level Classification

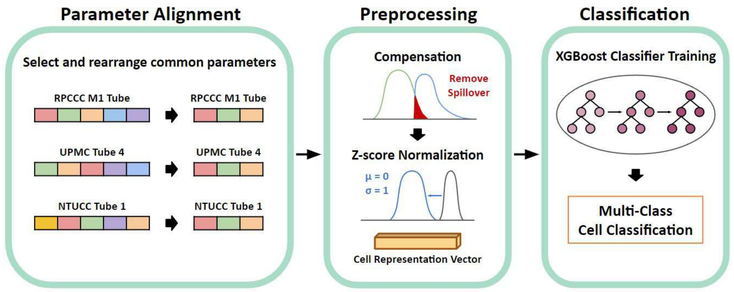
Cell Level Dataset

A total of 53 newly-diagnostic non-neoplastic FC records were collected from three sites (one from Taiwan and two from US). Each laboratory applied different flow panels. We selected one tube from each panel for analysis. 8 parameters were found commonly adapted in these tubes.

Dataset	RPCCC M1 Tube	UPMC Tube 4	NTUCC Tube 1	All Sites
Flow Panel	ClearLLab10C	UPMC Diagnostic	Euroflow AML/MDS	3 panels
Sample Number	21	22	10	53
Parameters (n)	14	14	11	8 in common**
Total Cell Number (n, ratio)	1,480,519	659,952	9,999,993	12,140,464
Granulocyte	569,711 (38.48%)	377,267 (57.17%)	4,133,242 (41.33%)	5,080,220 (41.85%)
Lymphocyte	179,841 (12.15%)	96,706 (14.65%)	18,191 (0.18%)	294,738 (2.43%)
Monocyte	29,680 (2.00%)	14,323 (2.17%)	84,365 (0.84%)	128,368 (1.06%)
Blast	14,094 (0.95%)	5,749 (0.87%)	19,577 (0.20%)	39,420 (0.32%)
Others	687,193 (46.42%)	165,907 (25.14%)	5,744,618 (57.45%)	6,597,718 (54.34%)

** Common parameters: FSC-A, FSC-H, SSC-A, CD11b, CD13, CD16, CD45, HLA-DR

Cell Level Classification Framework



1. Select and rearrange common parameters across different panels.
2. Pre-process the fcs files with compensation and site-wise z-score normalization
3. Apply 5-fold cross validation on sample-level to train and validate an eXtreme Gradient Boosting (XGBoost) classifier for multi-class classification. Training data is sampled and balanced before classifier training.

Cell Level Classification Results

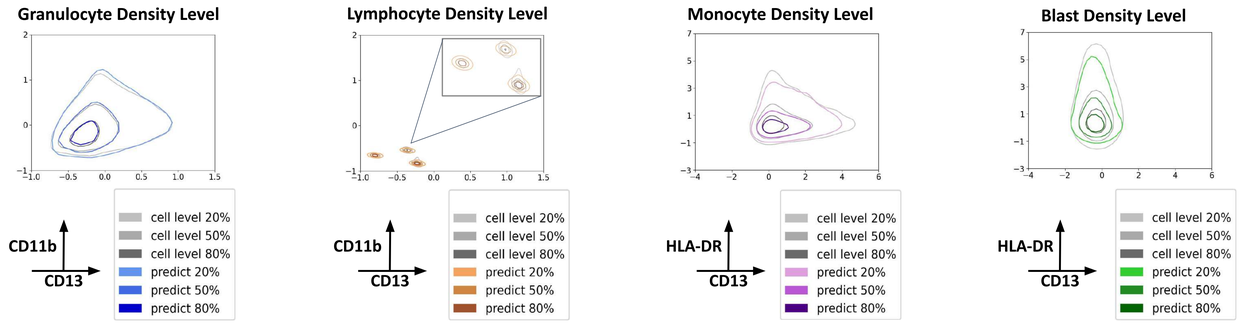
The cell level classification models trained with 8 common parameters achieved comparable performance with an AUC of 97.10% compared to models trained with site-dependent parameters. (AUC ranging from 97.60% to 99.70%)

Dataset	RPCCC	UPMC	NTUCC	RPCCC	UPMC	NTUCC	All Sites
Parameters (n)	14	14	11	8 common parameters			
AUC	99.70%	98.50%	97.60%	99.50%	98.20%	96.40%	**97.10%**
Accuracy	95.50%	92.70%	92.20%	94.20%	90.60%	88.60%	**88.30%**
UAR	96.70%	94.30%	95.00%	95.90%	91.90%	94.50%	**93.90%**
Weighted F1	95.70%	92.80%	92.30%	94.60%	91.00%	89.80%	**89.10%**

Abbreviation: AUC: Area Under the receiver operating characteristic (ROC) Curve, UAR: Unweighted Average Recall

Density Contour Comparison between Cell Label and Cell Prediction of All Sites

Density Contour Plot is a visulization tool that applies Kernel Distribution Estimation (KDE) to approximate the Probability Density Function (PDF) of given data, where each curve line represents a certain level of PDF. For example, the 20% level curve means that 20% of the given data points lie **outside** the line. In this experiment, both cell label and cell prediction are used for visualization. The closer the corresponding label curve and prediction curve are, the preciser the cell classification is.



Conclusion

- This work shows that our approaches are able to discover and distinguish panel-agnostic characteristics in both sample level and cell level.
- The results demonstrate the potential of automatic ML pipeline to enhance the consistency, capability, and capacity of flow cytometry analyses.
- Future Work: Investigate further in discordance cases to improve model performance, and expand dataset to cover wider spectrum of panels and more instrument models.

Suppl. Data

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